Kaustubh Hemant Sonawane

Bristol, UK | ksstubh.sonawane@gmail.com | +44 7388229743 | +91 8591441019 | www.github.com/Arishtanemi4 | www.linkedin.com/in/kaustubhson4 | https://kaustubh-sonawane-portfolio.vercel.app

UK Visa Status: Student Visa. Eligible for Graduate Visa (2 Years Full Working Rights)

17 October 2026

Hiring Team
Aventum Group

Application: AI Engineer (VN583)

Dear Aventum hiring team,

I am applying for the AI Engineer role (VN583). Over three years at HDFC Asset Management I built and deployed LLM applications, agentic tool-calling workflows and RAG services used by the Compliance, Client Service and Investment teams, taking each one from requirements through release and then improving it on user feedback. Your posting asks for AI that runs inside real products rather than in notebooks, and that is the kind of work I have done.

At HDFC I designed and built these applications end to end: a ReactJS frontend, Python FastAPI microservices packaged in Docker and deployed on AWS through a one-click Jenkins and GitLab CI/CD pipeline, LlamaIndex with Azure OpenAI and a self-hosted Llama-3, and a separate pgvector database per department. Agents called Python functions as tools to run SQL-backed fund comparisons. I also built a Python pipeline, scheduled with EventBridge, Lambda and SageMaker, that parses over 300 rating-agency emails a day into structured data and saves 60 analyst hours a month. Data pipelines, inference services and integration through APIs and microservices are familiar ground.

My sequence-modelling work is academic. In the MSc VisionCaptioner team project we compared CNN, ViT and ResNet encoders with symbolic, embedding-retrieval and LSTM decoders, with and without attention, across five seeds. My MSc dissertation with AstraZeneca, CellLineSelector, is a scoring and ranking tool; my focus was the data preprocessing, the scoring algorithm and the RAG layer, and I refined it with feedback from AstraZeneca scientists.

I want to be direct about where I am less experienced. I have not built behavioural models: no imitation learning or behavioural cloning, no Markov or decision-process models and no Digital Twins. I have not used causal inference tooling such as DoWhy, EconML or CausalML, and I have not validated a deployed model with A/B tests, shadow deployments or offline replay. I cannot point to a production model built in PyTorch or TensorFlow; my production work was Python services and GenAI, and my deep learning is from my degrees. At HDFC quality came from user feedback, system-prompt guardrails and Compliance review rather than evaluation harnesses or monitoring dashboards. These are the areas where I would expect to grow fastest.

What I can offer is ownership in an engineering-led environment. The first GenAI systems at HDFC were a two-person effort between my manager and me, I worked directly with business teams to turn their requirements into tools they could use, and in my final year I acted as de facto team lead for up to four interns. Python is my day-to-day language, and two of my projects were recognised as Best Team Project (2024, 2025).

I am currently based in Bristol, hold a Student Visa and am eligible for the Graduate Visa, which gives two years of full working rights in the UK. The posting is for London with a hybrid model, so I would be glad to discuss the working arrangement. Thank you for considering my application; I would be glad to talk it through.

Yours sincerely,
Kaustubh Sonawane